Lesson 01
Bipolar Junction Transistors

Interactive Seminar

SCHOOL OF ENGINEERING

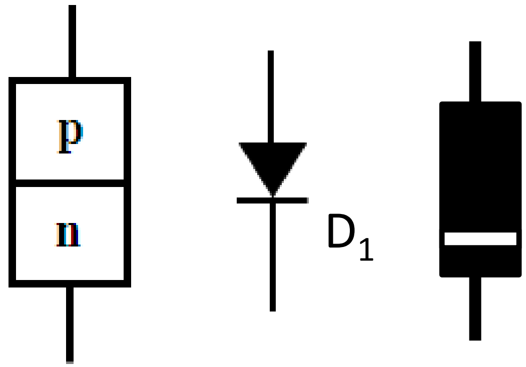
E219
Analogue Electronics

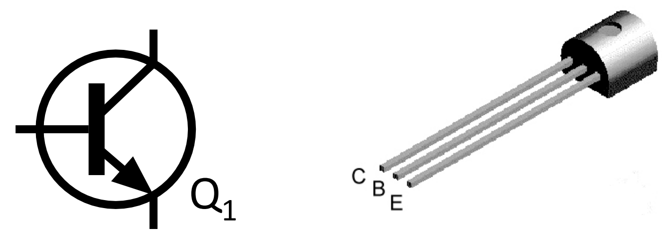
Diode & Bipolar Junction Transistor

Refer to Smartbook Sec 29-1. Read up the section and fill up the blanks below:

- Diode is a semiconductor device that made up of a single PN junction

- Diode has 2 terminals, named as **(A)node**, & **(K)Cathode**





- Is Bipolar Junction transistor (BJT) a semiconductor device?

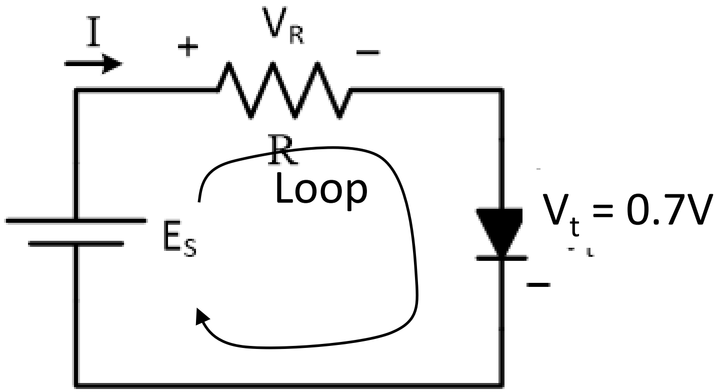
- How many PN junction does BJT has?

- Why is Q_1 a NPN type BJT?

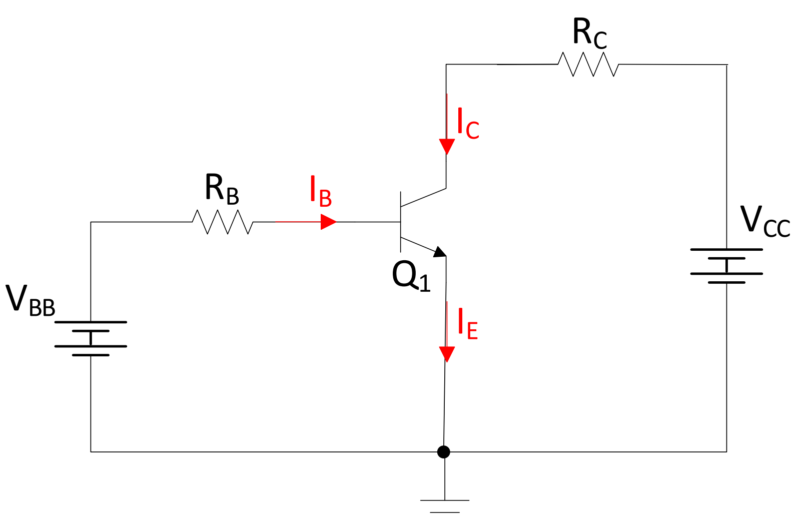
- How is the 3 BJT terminals label as Base**(B)**, Collector**(C)** & Emitter **(E)**?

Biasing a NPN BJT

Refer to Smartbook Sec 29-2



Diode Bias Circuit



BJT Bias Circuit

- External Voltage Source to bias a Diode circuit
- $E_S = V_R + V_t$
  $E_S > V_t$
- What is the difference between BJT biased Circuit and a Diode biased circuit?
- What is the difference between Figure 29-4 of the smartbook and the circuit shown on the left?
- What is the purpose of having a V_{BB}?

Important: All currents shown in Smartbook is <u>electron flow</u> while the currents used in E219 is <u>conventional flow</u>

Biasing a NPN BJT

Refer to Smartbook Sec 29-2, Figure 29-5,

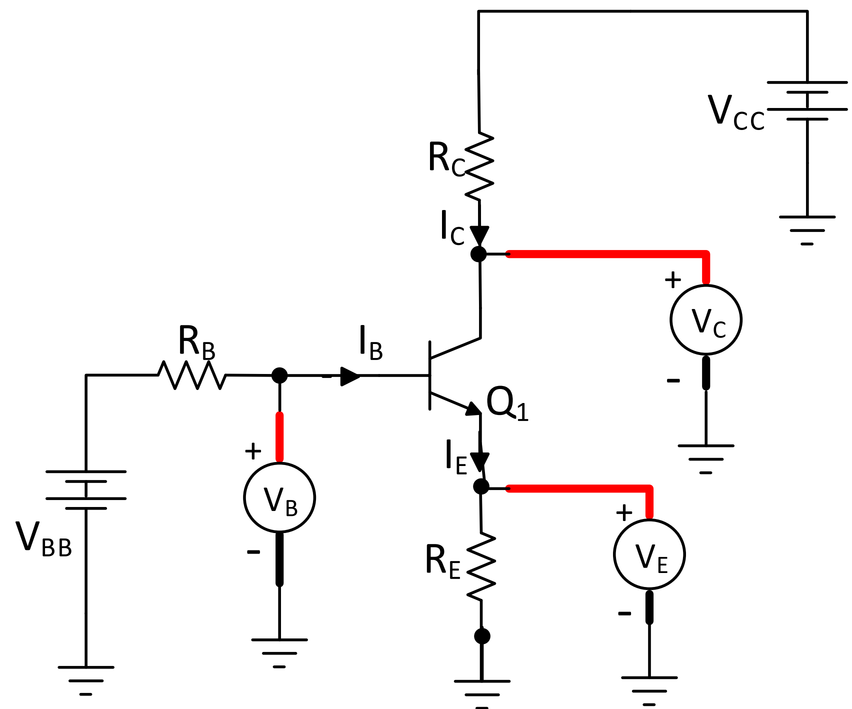
External Voltage Source(s) is needed to provide currents through different parts of the BJT

External DC sources

- V_{BB} : Base Supply Voltage
- V_{CC} : Collector Supply Voltage

Node voltages at BJT terminals

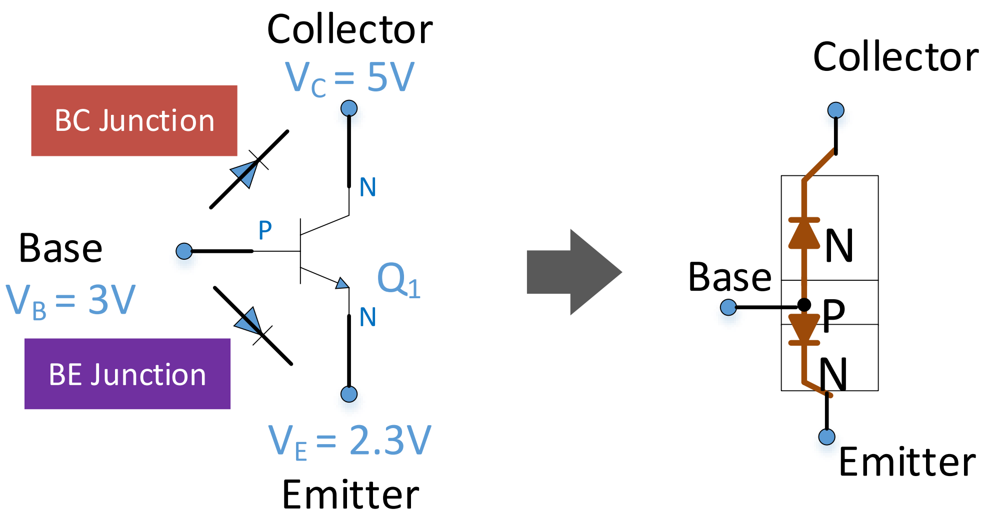
- V_B : Node Voltage at Base
- V_C : Node Voltage at Collector
- V_E : Node Voltage at Emitter



Important Note: ***Node Voltage** is measured voltage between the given node and ground*

BJT Voltages

Calculation of voltage across two terminals using node voltages



A transistor behaves like two back-to-back diodes : one between terminal B and E, another between terminal B and C.

Typically Forward Bias in a Junction diode ≈ ***0.7V*** *unless stated otherwise*

Voltage across BE terminal

$$V_{BE} = V_B - V_E$$

$V_{BE} =$

Voltage across BC terminal

$$V_{BC} = V_B - V_C$$

$V_{BC} =$

Voltage across CE terminal

$$V_{CE} = V_C - V_E$$

$V_{CE} =$

Circle the appropriate ***Bias Condition*** *for the respective junction diodes*

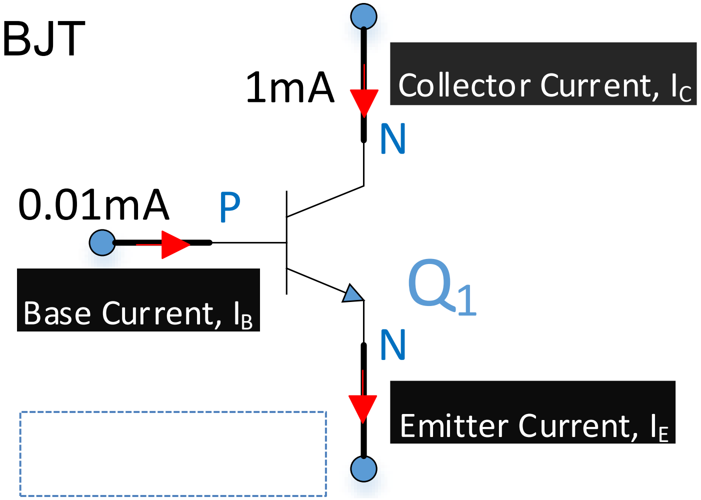
BE Junction : Forward / Reverse Bias

BC Junction : Forward / Reverse Bias

BJT Currents

Refer to Smartbook Sec 29-2

- If BE junction is forward biased, currents flow in BJT
  - ✓ I_B is very small
  - ✓ $I_C \approx I_E$
- Based on conventional current flow



I_E =

<u>By Kirchoff Curent Law (KCL):</u>

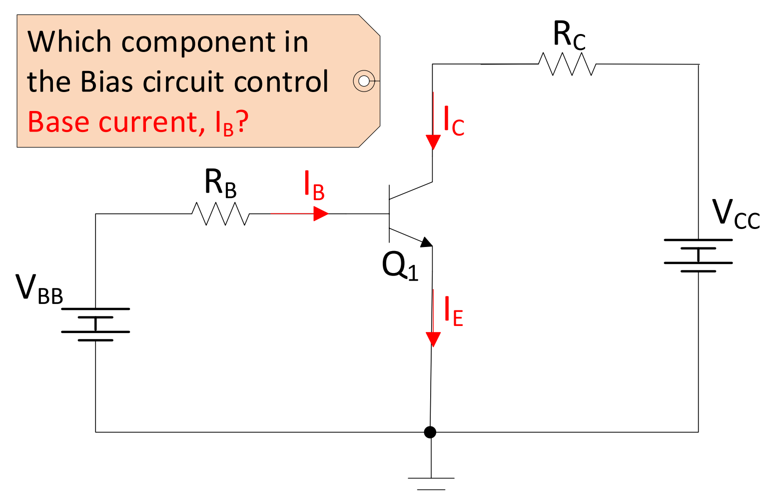
$$\boxed{I_E = I_B + I_C}$$

- **DC beta,** $\boxed{\beta_{DC} = I_c / I_B}$

DC beta, β_{DC} for Q_1 =

- **DC alpha, α_{DC}** is the ratio of I_C to I_E

DC alpha, α_{DC} for Q_1=

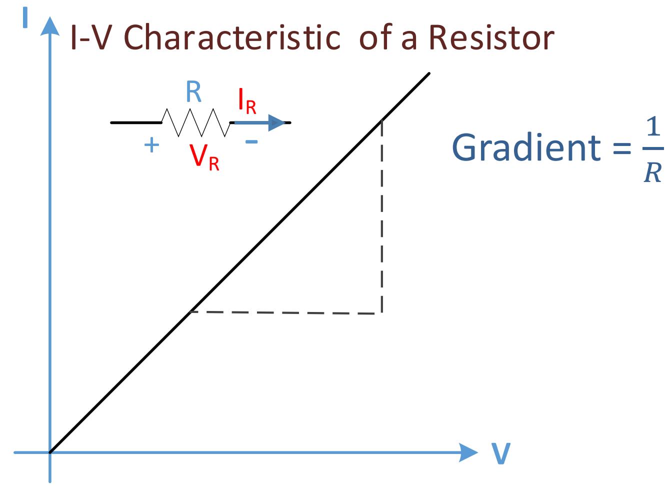


<u>BJT Bias Circuit</u>

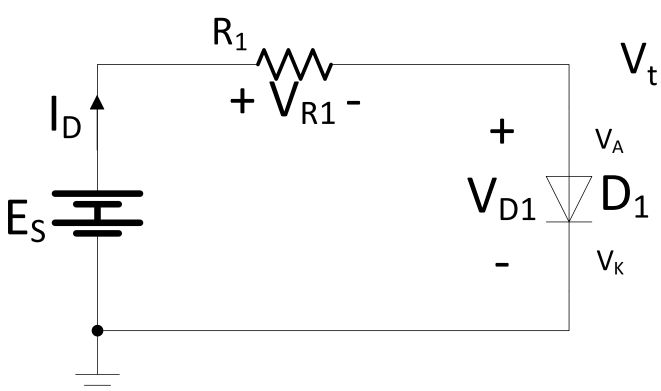
BJT - A Non-Linear Device

Recap I-V characteristics of Resistor and Diode

- Resistor is a **linear** device.
- For a fixed R, $V_R \propto I_R$

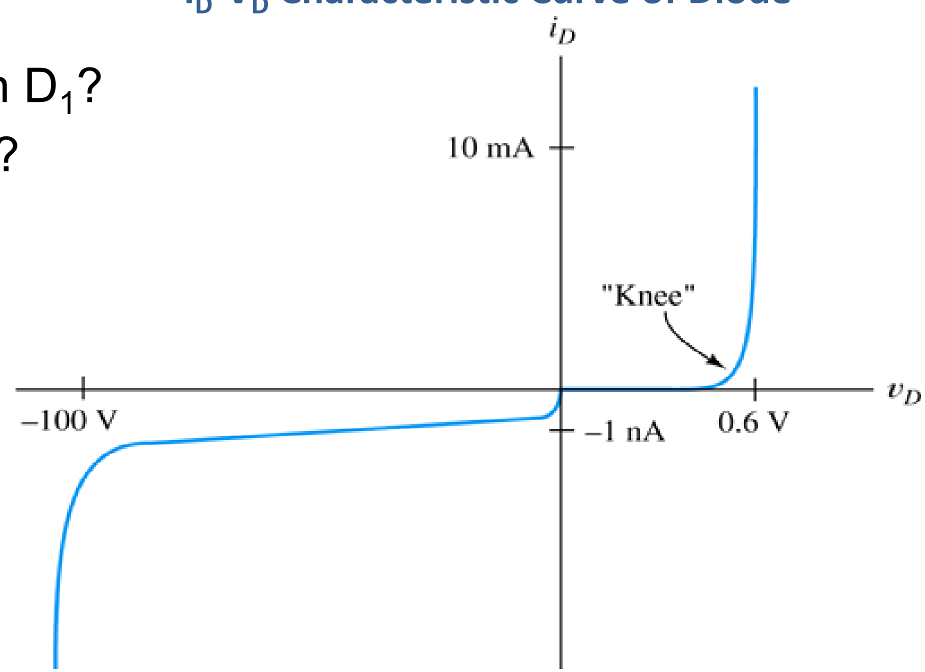


- Is Diode a **linear** device?
- When does current starts to flow through D_1?
- Which component limits the current flow?



V_t = 0.7V

I_D-V_D Characteristic Curve of Diode

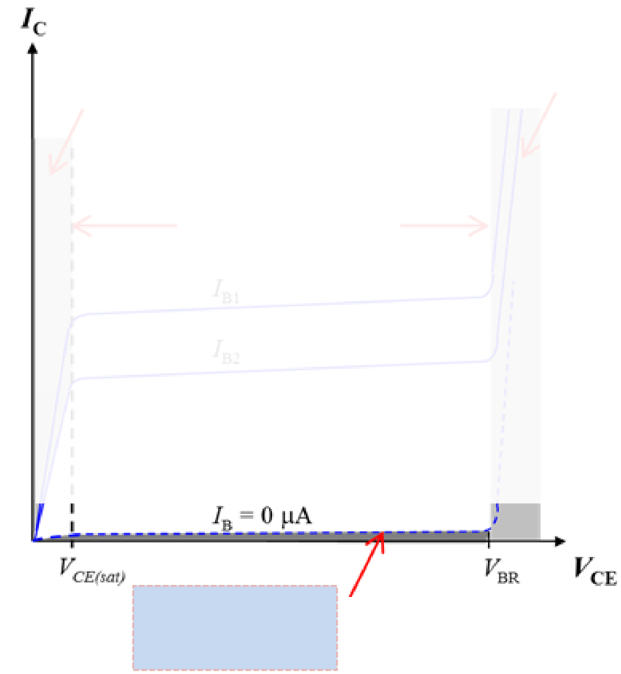


BJT Operation mode : Cut-off Mode

Refer to Smartbook Sec 29-3

- BJT is “fully off”
  - ✓ I_B is very small
  - ✓ $I_C \approx 0A$
- Terminal C-E behaves like Open Circuit

BJT Operation Mode	B-E Junction	B-C Junction
Cut-off	Reverse Bias	Reverse Bias
Active	Forward Bias	Reverse Bias
Saturation	Forward Bias	Forward Bias



BC Junction is

Biased

$V_{BC} = \quad V_B \quad - \quad V_C$

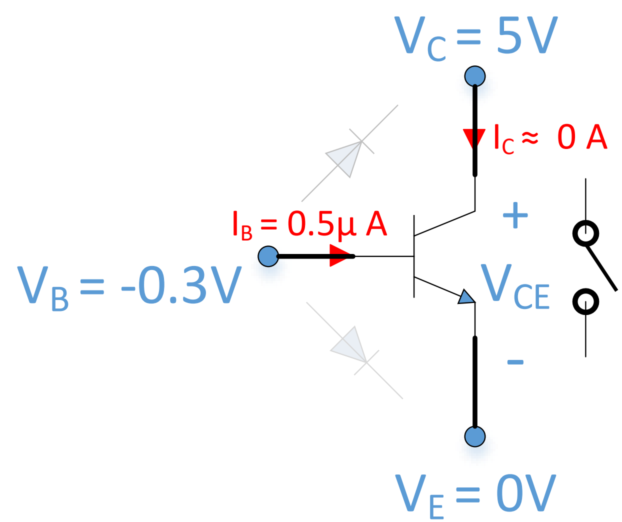
$V_{BC} =$

BE Junction is

Biased

$V_{BE} = \quad V_B \quad - \quad V_E$

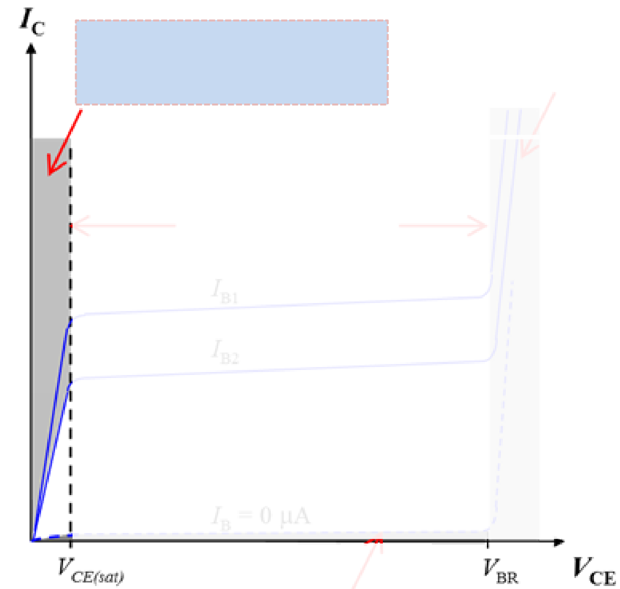
$V_{BE} =$



BJT Operation mode : Saturation Mode

Refer to Smartbook Sec 29-3

- BJT is "fully on"
  - ✓ $\beta_{DC} \neq \frac{I_C}{I_B}$
  - ✓ $V_{CE} \leq 0.2V$
- Terminal C-E behaves like Short Circuit



BJT Operation Mode	B-E Junction	B-C Junction
Cut-off	Reverse Bias	Reverse Bias
Active	Forward Bias	Reverse Bias
Saturation	Forward Bias	Forward Bias

BC Junction is Biased

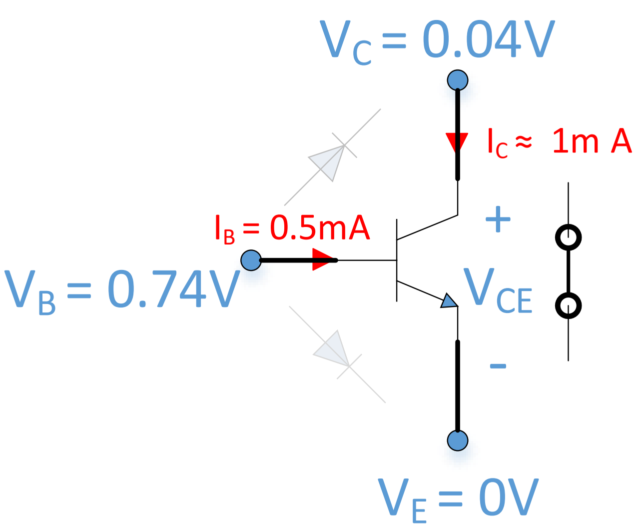
$V_{BC} = V_B - V_C$

$V_{BC} =$

BE Junction is Biased

$V_{BE} = V_B - V_E$

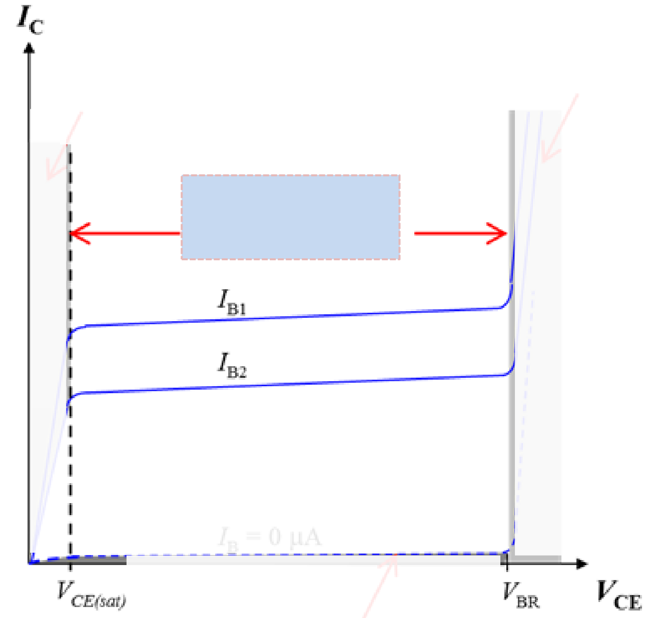
$V_{BE} =$



BJT Operation mode : Active Mode

Refer to Smartbook 29-3

- BJT is "Partially on"
  - ✓ $I_C = \beta_{DC} \times I_B$
  - ✓ β_{DC} remains constant regardless of different I_B



BJT Operation Mode	B-E Junction	B-C Junction
Cut-off	Reverse Bias	Reverse Bias
Active	Forward Bias	Reverse Bias
Saturation	Forward Bias	Forward Bias

BC Junction is Biased

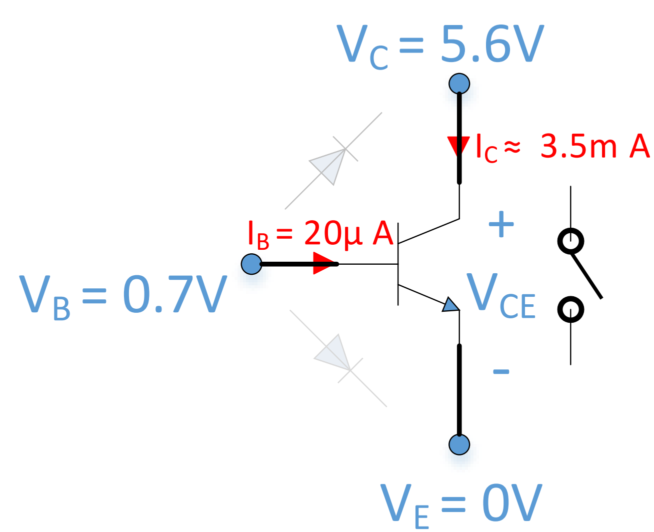
$V_{BC} = V_B - V_C$

$V_{BC} =$

BE Junction is Biased

$V_{BE} = V_B - V_E$

$V_{BE} =$



Practice 1

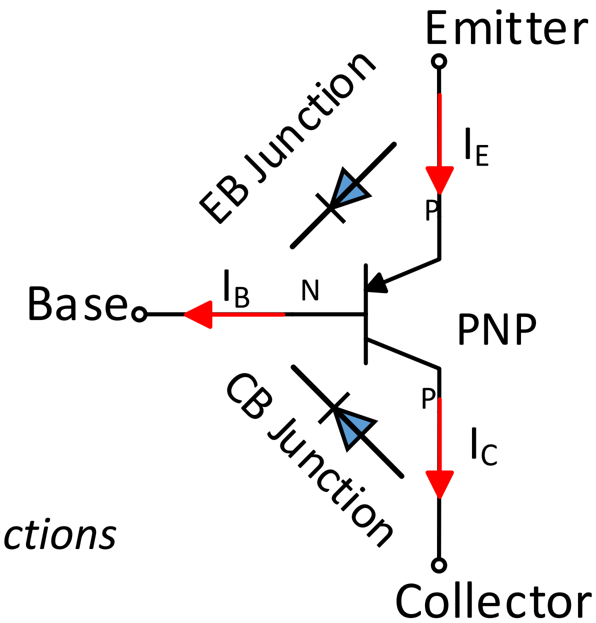
PNP BJT Currents and Voltages

PNP is the complement of NPN BJT

KCL of BJT current remains the same **except** that the **current direction is opposite**

By KCL:

$$\boxed{I_E = I_B + I_C}$$



Take note that the **2 PN junctions of PNP are reverse**

In NPN, the 2 PN junctions are BE-Junctions and BC junctions

Circle the appropriate bias condition for different operating modes of a PNP

BJT Mode Of Operation	E-B Junction	C-B Junction
Cut-off	Forward / Reverse Bias	Forward / Reverse Bias
Active	Forward / Reverse Bias	Forward / Reverse Bias
Saturation	Forward / Reverse Bias	Forward / Reverse Bias

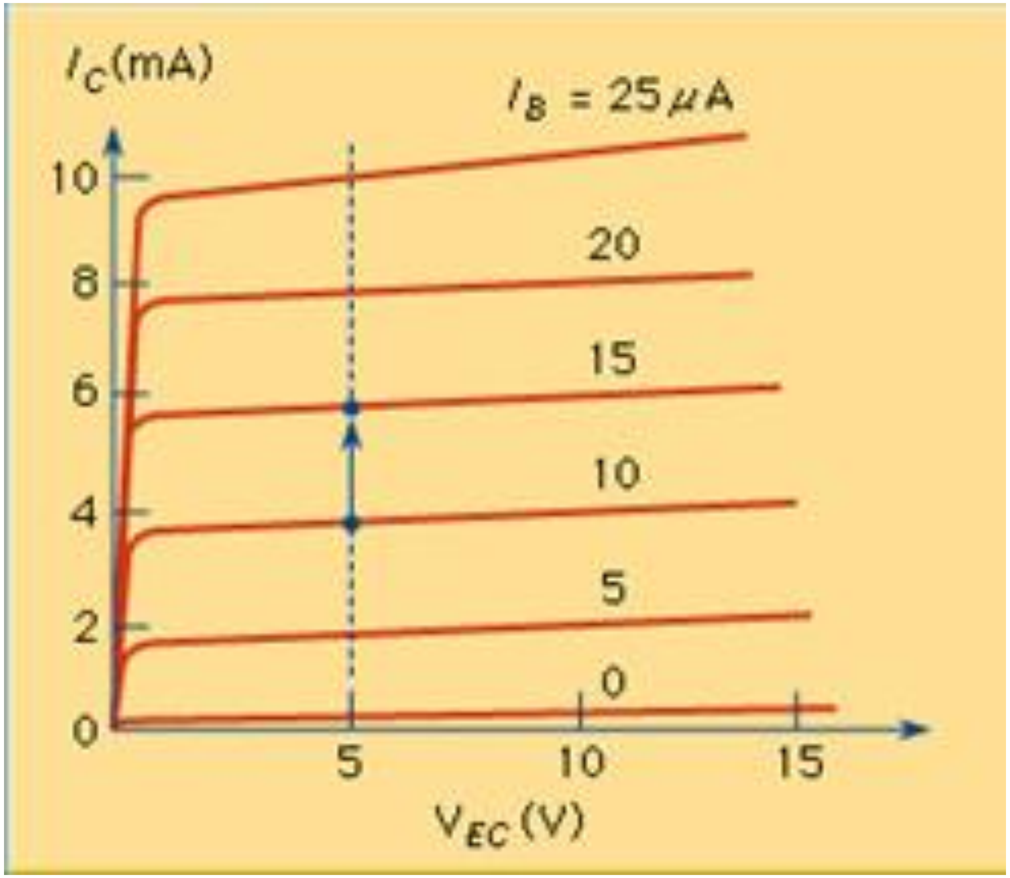
Note: *there are no difference in the bias condition for the 3 operation mode irregardless of PNP or NPN*

I_C-V_{CE} Characteristic Curve of PNP

- Similar to NPN, small increase in I_B of a PNP generates a large increase in I_C during active operating mode :

$$I_C = \beta_{DC} I_B$$

Is β_{DC} value the same in ***Active*** **and** ***Saturation*** **operating Mode?**



I_C-V_{EC} Characteristic Curve of PNP

Practice 2

Learning Outcomes

- Describe the basic structure of a BJT
- Differentiate between two types of BJT: PNP; NPN and recognize the electrical symbol of each BJT
- Describe the I-V characteristics and parameters for a BJT
- Explain the various BJT operation region/mode:
  Cut-off; Active; Saturation
- Explain the effect on the junction currents IB, IC, IE and junction voltages VB, VC by external parameters such as VBB, VCC, RB, RC.

Useful References

1) Transistors, how do they work?

   https://www.youtube.com/watch?v=7ukDKVHnac4&t=3s

2) What is transistor?

   https://www.linkedin.com/learning/electronics-foundations-semiconductor-devices/what-is-a-transistor?u=41614228